UNIT42 · FIRMWAREX

Firmware Security Assessment

Sample firmware

Generated 10/19/2026, 10:57:00 PM

Executive summary

This assessment recorded **17 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	10	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	x86_64	Present	Not detected	partial	Present	Not detected
/bin/taint-demo	x86_64	Present	Not detected	partial	Not detected	Not detected

Detailed findings

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1. Memory safety candidate High

Property	Result
CWE	CWE-787
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	probable
Review status	new
Address	0x500008
Function / sink	strcpy
Detected	2026-10-19T17:25:57.037221Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x401136 → 0x401040 → 0x500010 → 0x40115c → 0x40116b → 0x401030 → 0x500008
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract" … [excerpt; full expression in JSON export]
fault address	0x404030
overflow offset	8
stack hash	bfc322b46d82910dab6757cff82dd695801b2df2172325cb b73b4e88d0926fc2

Evidence reference: evidence:28292280e79263240d709255f46c3e921567eaf038d13eb02eb8c4aa6cb8746c. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

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2. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established

Property	Result
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134182Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	8dff21febc71e670d6e79611840f9e3bb0637b13d714ac0a98c5f8354e81165f
byte offset	237
byte length	3

Evidence reference: evidence:3ac3d0c5a7a9ed97132519794fbc5af0eadd1cca187d00a1d1db151b7b6e7d83. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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3. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134202Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	7ed55c8380043ef0f67a228aaed18350dea4172137e444b54c8a4ff6f0764b8c
byte offset	255
byte length	3

Evidence reference: evidence:3e797f294f6e88bf6459aaabc9fb3095e784b8df76cb9d38c93b70fcd66403fc. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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4. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134034Z

Evidence

Observation	Value
rule	credential

Observation	Value
fingerprint	f4da6c6c76f447d53186b3fea55b7d4e9538c8e8b7917d500464b14748273e6b
byte offset	9
byte length	19

Evidence reference: evidence:c8b83ab61708035b4660d144ff6a4d79c50c6db1bc35ba6b0db1f4563a6575df. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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5. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134085Z

Evidence

Observation	Value
rule	credential
fingerprint	3f182cb32cbe353e3f86e9be69156a986933b759a34db1add10e00e16878e758

Observation	Value
byte offset	37
byte length	24

Evidence reference: evidence:cc66bad63828e88834011aec265f8e48817227a0f2c008fe94b582200b29a919. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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6. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134109Z

Evidence

Observation	Value
rule	private_key
fingerprint	9952429680c1a260852f849647eacd33365cb066705dcbfd425c1d246074fc7c
byte offset	62
byte length	78

Evidence reference: evidence:b1eda05b8a555f4b23d6f6f9740a0acb1a5ef15381b074a68d8ef61a08ff87b0. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-4fd43bfb7a09fe86
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134360Z

Evidence

Observation	Value
rule	credential
fingerprint	a7e9701e281673960eb1842ef11c4c18c6571a493e26bbd58fb185a3a92cf943
byte offset	38
byte length	25

Evidence reference: evidence:2051e38e55601e4d4c677228bd00db56aa875c4de8e58f38159fcfba96b890ee. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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8. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134164Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	bc8a723f0ea035afb5e8fa072c46d1ccdc63f93c3d9a61bca45cdd43ff6de83c
byte offset	259
byte length	29

Evidence reference: evidence:ed77d16b0d98aa234c03247a6d2ca784b1ebb19f2fb12824badb4e7db8ce15f9. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

9. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-19T17:25:52.400860Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

10. Memory safety candidate Low

Property	Result
CWE	CWE-676

Property	Result
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-19T17:25:52.401123Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

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11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic

Property	Result
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.133015Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	eb5417f3d2d6ae09e8d4b01098c94152df71cb329a79c6dd20ce49183eb0260f
byte offset	511650
byte length	50

Evidence reference: evidence:b4ab09de55549d726a9fc995108007b8da93e382c1cba694bdef98c5d5e68804. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established

Property	Result
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.133099Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	15c1bcad9cbb58021cf6c918bf3f9bdfcb6fe1d2a23fee62625eb56f8b3b002b
byte offset	591137
byte length	26

Evidence reference: evidence:cc706f9c1812307cc4dfc72bc767beeb43feaa092d7d1eb54c4dafd11badd4d3. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.133127Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	6eae61841643b51e0f63a0a2d7d832a8e56cef1ea17f003779a5c8009c12c92d
byte offset	591169
byte length	26

Evidence reference: evidence:2a4bf4375ff96aba81e2abf5e9a212f9aadd4f4574194a575a5aea75b53b622f. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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14. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.133149Z

Evidence

Observation	Value
rule	high_entropy

Observation	Value
fingerprint	a28e9d63b01fa1038ba40571436488b1aff594cf78f46526ecddba5a583abff6
byte offset	591328
byte length	36

Evidence reference: evidence:bb2bc213e3adc0179565df77a03a34c7283ee6128cca74ca87e662804ff397dc. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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15. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.133168Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	818993f7f9c57414bd94ac03d6038fcbb9b2016d14f2eecdd07e861c4e1fcef1

Observation	Value
byte offset	591392
byte length	36

Evidence reference: evidence:77b91322cde040b1ba1fed5aa32959e4786215f0a1f7951b74c2c3e934c23751. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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16. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134128Z

Evidence

Observation	Value
rule	certificate
fingerprint	c2a8206a29a7e7c7c9f3cd687bc48030460e75b50b20b845c6ae07257be3d4d4
byte offset	141
byte length	75

Evidence reference: evidence:78ce785526be29cad8f9fbc0fbbbc44a2eee42717a95331941e83ae680cc8705. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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17. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-19T17:25:57.134147Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	7c91417468e96cc83fa2e6d56d9be45bef0373c2edcabc6220907e496b811527
byte offset	29
byte length	32

Evidence reference: evidence:00e41a5f108b01dd8ad8454878751b8bb5c8d34f3c9804ef610ae23d57c0669a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	54f9ba27-7a3a-455a-93cb-e694a1406a36
SHA-256	a71d1e4937d2eb468ea965462c27f6b3c99eb2207fa39bd59d04161bf45856c9
Job ID	0d176e7f-7ceb-43bc-96f9-feda0942ea6e
Image bytes	339968